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# Map interface in Java

## => Abstract Design Pattern :

-> It provides a way by which we dont have to implement all the methods of an interface, we just have to declare an abstract class which inherits the interface and in that abstract class we will override the methods of an interface. When we create a class which inherits the abstract class, then we dont need to override all the methods of interface

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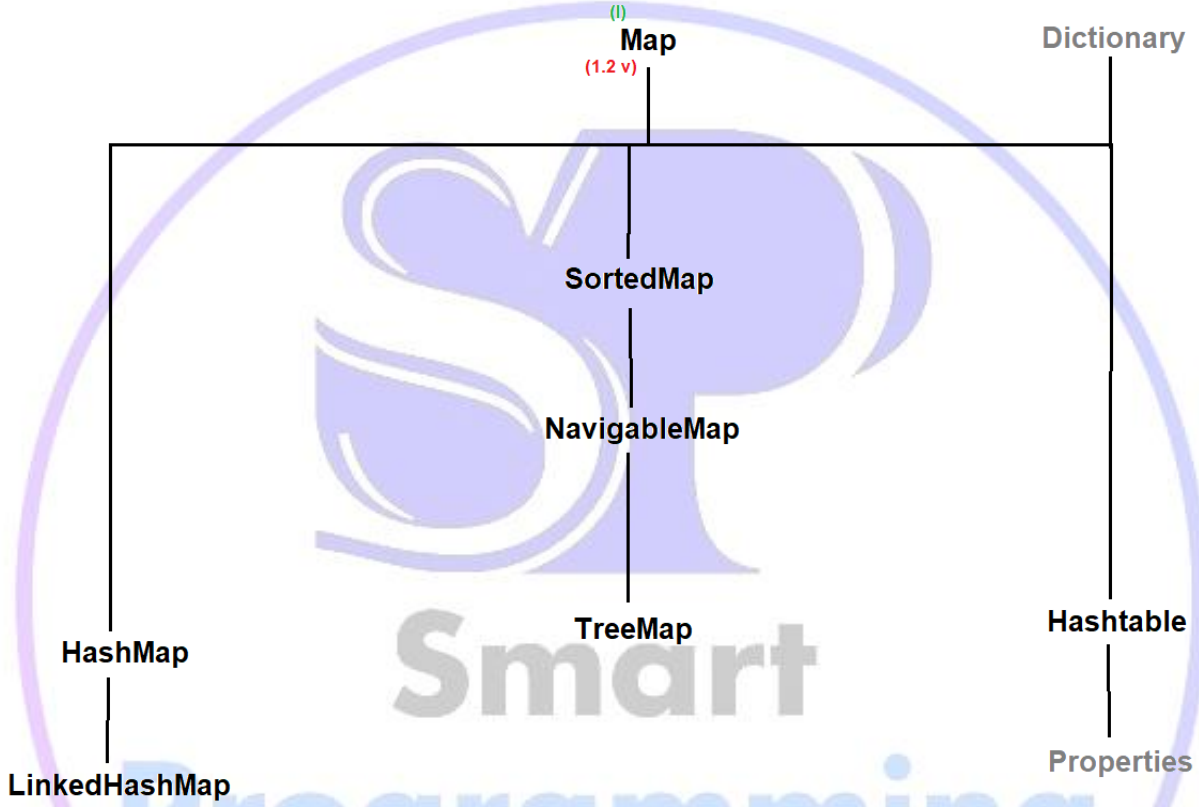
## => Map :-

- ➔ Map is an interface which is present in java.util package
- ➔ Map does not inherit Collection interface
- ➔ Syntax : public interface Map { - }
- ➔ Map was introduced in JDK 1.2 version

## ➔ Hierarchy of Map interface :-




## ➔ Properties of Map :-

1. Map stores the data in key-value pair. Each key-value pair is known as Entry
2. In Map, keys should always unique but values can be duplicate
3. Map can store heterogeneous elements or different type of elements

4. In Map keys we can store maximum one null value but in values we can store any number of null values
5. Map does not follows the insertion order by default
6. Map does not follows the sorting order by default

**➔ Methods of Map interface :**

1. public Object put(Object key, Object value)
2. public void putAll(Map map)
3. public Object get(Object key)
4. public Object remove(Object key)
5. public void clear()
6. public boolean isEmpty()
7. public int size()
8. public boolean containsKey(Object key)
9. public boolean containsValue(Object value)
10. public Set keySet();
11. public Collection values();

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